ECO 362 Financial Investments
Midterm Exam

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Instructions

1. Check that you have all 4 pages. There are 4 questions.

Question	Points
1	25
2	25
3	20
4	30
Total	100

2. This is a closed book exam. However, you may use a one-page cheat sheet containing formulas or anything else you may find helpful. The cheat sheet must be letter size (8.5x11in) with writing on one side only (i.e., the back side must be left blank). You are under the Honor Code to comply with this requirement.

3. You may use a writing instrument, a cheat sheet, and a calculator. Put everything else away. The calculator cannot be a phone, a tablet, or a laptop. If you do not have a calculator, you may borrow one. Please return it after the exam.

4. You may not communicate with other students or violate the Honor Code during this exam. Write your name and sign the Honor Code on the first page of the exam book.

   I pledge my honor that I have not violated the Honor Code during this examination.

5. Write your answers in the exam books. *You must show all steps necessary in deriving your solution for any credit.* Report the yield, the coupon rate, or the return in percentage points. Round all answers to two decimal places (e.g., $95.67 or 1.23%).

6. You have 90 minutes to complete the exam. After 90 minutes, you must stop working on the exam. You then have an extra 20 minutes to submit your answers as one scanned pdf file or photographed images through Gradescope. If you do not have a device to scan or photograph, give your exam books to a preceptor, who will be able to do it for you.

7. In addition to submission by Gradescope, submit the hard copies of the exam books before you leave the room. We need them in case there are any technical issues (e.g., unreadable scans). Make sure that your name is on the exam books that you submit.

1. The following table reports annually compounded par yields in percentage points in 2023. Assume no arbitrage.

Maturity (years)	Yield (%)
1	4.87
2	4.36
3	4.12

[25 points total: 5 points each part]

(a) In 2023, what is the price of a 2-year zero coupon bond that pays \$100 in 2025?

(b) In 2023, what is the price of a 3-year zero coupon bond that pays \$100 in 2026?

(c) In 2023, what is the price of an asset that pays \$50 in 2025 and \$200 in 2026?

(d) In 2023, what is the continuously compounded forward rate on a loan from 2024 to 2026?

(e) In 2023, you buy insurance that pays \$100,000 if your house floods in 2024. The probability of a flood is 2%. What is the price of flood insurance?

2. There are two possible states in years 1 and 2: expansion or recession. The probability of an expansion is 0.8 and a recession is 0.2 in both years. The following table reports the value of the stochastic discount factor for each future date and state.

	Future state	
Year	Expansion	Recession
1	0.58	2.60
2	0.55	2.50

The following table reports the cash flows of two risky assets for each future date and state. Assume no arbitrage.

	State in year 1		State in year 2	
Asset	Expansion	Recession	Expansion	Recession
A	\$110	\$80	\$115	\$85
B	\$100	\$90	\$0	\$0

[25 points total: 5 points each part]

(a) What is the price of a 1-year zero coupon bond per \$1 face value?

(b) What is the annually compounded yield on a 2-year zero coupon bond?

(c) What is the price of asset A?

(d) What is the price of asset B?

(e) What is the annually compounded expected return on asset B?

3. For each part, write "Arbitrage" if there is an arbitrage opportunity, or "No arbitrage" if there is no apparent arbitrage opportunity based on the information given. If you write "Arbitrage", you must write the quantities of the assets that you would go long/short for full credit. For example, you must write "Long 1 unit of asset A and short 1 unit of asset B". If you write "No arbitrage", you do not need to provide further justification for your answer.

   [**20 points total: 5 points each part**]

   (a) Bonds A and B are both 5-year coupon bonds with face value of \$100 and price of \$95. Bond A has a coupon rate of 2%, and bond B has a coupon rate of 1%.

   (b) Bonds C and D are both 10-year coupon bonds with face value of \$100. Bond C has a coupon rate of 1% and an annually compounded yield of 1%. Bond D has a coupon rate of 2% and an annually compounded yield of 2%.

   (c) Bonds E and F are both coupon bonds with face value of \$100 and a coupon rate of 2%. Bond E has a maturity of 5 years and an annually compounded yield of 2%. Bond F has a maturity of 10 years and an annually compounded yield of 3%.

   (d) The following table contains the current price and future payoffs of three state-contingent assets.

		Payoff	
Asset	Price	State 1	State 2
A	0.99	1.00	1.00
B	0.05	0	0.10
C	0.10	0.20	0

4. The following table reports the mean and standard deviation of returns for three assets. The riskless asset has a constant return. The correlation between USA and Canada is zero.

Asset	Mean (%)	Standard deviation (%)
Riskless	0.08	0
USA	0.87	4.30
Canada	0.89	5.65

Assume that the investor has mean-variance expected utility:

$$\mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p),$$

where R_p is the portfolio return.

[30 points total: 6 points each part]

(a) Compute the portfolio weights for the tangency portfolio. Report the portfolio weights for all three assets.

(b) What is the Sharpe ratio for the tangency portfolio? Recall that the Sharpe ratio is the ratio of mean excess returns to the standard deviation of returns:

$$\frac{\mathbb{E}[R_p - R_b]}{\sigma(R_p)},$$

where R_p is the portfolio return and R_b is the riskless interest rate.

(c) Compute the portfolio weights for the minimum variance portfolio. Report the portfolio weights for USA and Canada.

(d) Portfolio A is 31% in USA, 19% in Canada, and 50% in the riskless asset. Portfolio B is 25% in USA, 22% in Canada, and 53% in the riskless asset. Which of the two portfolios is closer to an optimal portfolio for an investor, whose risk tolerance you do not know? You must provide a valid explanation for any credit (i.e., no credit for simply guessing A or B).

(e) Draw a figure of the efficient frontier and the capital allocation line. You must clearly mark the following points: riskless asset, USA, Canada, tangency portfolio, and minimum variance portfolio. Using this figure, explain why a portfolio that invests only in the USA is not efficient.